

Incident Response

Condensed Study Notes

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WEKK 5 INCIDENCE RESPONSE

Imagine your computer is like a busy office. Sometimes, unexpected things happen, like a printer jamming or a file not opening correctly. These are like 'incidents' in the office.

When an incident happens, there's a plan for what to do next, just like in an office where someone might call a technician or try to fix it themselves. This plan is called an 'incidence response'.

An incidence response is essentially a set of steps or actions taken when something goes wrong with a computer system. It's about handling problems efficiently to get things back to normal as quickly as possible.

Key ideas:

- An 'incident' is an unexpected problem or event that disrupts normal computer operations.
- An 'incidence response' is the organized way of dealing with these problems.

Introduction

We've already talked about 'incidents' (unexpected computer problems) and 'incidence response' (the plan to deal with them).

Now, let's dive a bit deeper into what 'incident response' actually means.

Think of 'incident response' like the specialized emergency response team that handles a specific type of crisis, say, a fire. They have their own gear, training, and procedures for putting out fires.

'Incident response' is a structured and planned way to find and deal with cyber attacks. The main goals are to:

- Reduce the harm caused by the attack.
- Get things back to normal as quickly as possible.
- Keep the overall costs down.

It's important to know that 'incident response' is actually a smaller part of a bigger picture called 'incident management'.

Imagine 'incident management' as the entire city's plan for dealing with any major disaster, like an earthquake. This plan involves many different groups of people – the mayor's office, lawyers, the HR department, the communication team, and the IT folks. They all have a role in the overall recovery.

'Incident response' is the specific piece of that big disaster plan that focuses *only* on the technical, cybersecurity tasks. It's like the firefighters (incident response) within the larger city disaster plan (incident management).

Types of security incidents

Before we dive into the different kinds of security problems, let's quickly connect a few ideas. Think of it like this:

- A **vulnerability** is like an unlocked window in your house – a weakness that someone could take advantage of.
- A **threat** is the person (or thing) that wants to take advantage of that unlocked window, like a burglar.
- An **incident** happens when the burglar actually gets through the unlocked window and into your house, or otherwise puts your belongings at risk.
- A **data breach** is a specific type of incident where the burglar successfully steals your sensitive stuff, like your family photos or your private journals.

It's always better to fix the unlocked window (the vulnerability) before a burglar (the threat) can use it to cause a problem (an incident) or steal your things (a data breach).

Now, let's look at some common ways these security problems can happen:

- **Unauthorized attempts to access systems or data:** This is like someone trying to open your doors or windows without permission, hoping to get inside your house or take things.
- **Insider threats:** This is when the problem comes from someone who already has permission to be in your house, like a roommate or family member who decides to steal something or cause damage.
- **Phishing attacks:** Imagine getting a fake letter in your mailbox that looks like it's from your bank, asking you to write down your account number and password. Phishing is a trick used online to get you to reveal sensitive information, like passwords or credit card details, by pretending to be someone trustworthy.
- **Privilege escalation attacks:** Think of your house having different levels of access. Maybe anyone can open the front door, but only you have the key to your bedroom. Privilege escalation is when an attacker who initially only has access to the front door figures out a way to get the key to your bedroom, gaining more control than they should have.
- **Password attacks:** This is like a burglar trying every key on a keychain, or guessing common combinations, to try and unlock your doors or a safe.
- **Man-in-the-middle attacks:** Imagine you're sending a secret message to a friend, but someone intercepts it, reads it, and then sends it on to your friend without either of you knowing. In this type of attack, the attacker secretly intercepts and possibly alters communication between two parties.
- **Malware attacks:** This is like someone introducing a bug or a virus into your house that damages things or spreads to other parts of your home. Malware (short for malicious software) is harmful code designed to damage or gain unauthorized access to computer systems.
- **Denial-of-service (DoS) attacks:** This is like someone blocking your front door with a huge crowd so that no one, not even you, can get in or out. A DoS attack aims to make a computer system or website unavailable to its intended users by overwhelming it with traffic.
- **Web application attacks:** These attacks target the websites and online services you use. It's like someone trying to find a weak spot in the design of a building's online storefront to get in or cause

problems.

- **Advanced persistent threats (APTs):** These are like a highly skilled and patient burglar who spends a long time planning and executing a break-in. They are sophisticated and ongoing attacks, often by well-funded groups, that aim to steal large amounts of data over a long period.

How to create an incident response plan

To handle computer problems effectively when they happen, you need to have a plan ready beforehand. This means thinking about what you'll do, checking if the plan makes sense, and even practicing it.

A good plan acts like a set of guidelines for the people who will be dealing with a computer problem. It helps them make smart choices when things get tough.

Here's how to build that plan:

1. **Create a Policy:** Think of this as the main rulebook. It's a document that outlines the overall goals and priorities for dealing with computer issues. It should be a living document, meaning it can be updated as needed.
2. **Build an Incident Response Team:** This is the group of people who will actually carry out the plan. You need to decide who does what and make sure everyone has the right training for their job. A plan is only as good as the people who execute it.
3. **Develop Playbooks:** If the policy is the rulebook, then playbooks are like detailed instruction manuals for specific situations. They describe the exact, step-by-step actions the team should take for different kinds of problems. This makes sure everyone responds in a consistent and efficient way, and it's also great for training new team members.
4. **Create a Communication Plan:** No plan can work without good communication. This plan outlines how the incident response team will talk to everyone involved. This could include other departments within the company (like executives, legal, or HR), customers, partners, and even the police or the public.

What is an incident response plan

Imagine your computer system is like a busy city. Sometimes, unexpected problems, like a traffic jam or a power outage (which we've called 'incidents'), can happen. An incident response plan is like the city's emergency preparedness manual.

This plan is a special set of documents that an organization creates to guide them when a security problem occurs. It's like a detailed map that tells everyone exactly what to do, step-by-step, from when a problem is first noticed until it's fixed.

This 'map' specifically answers key questions:

- **Who:** It clearly states which person or team is responsible for each specific task during an incident. It also explains how people can reach those responsible individuals.

- **What:** It defines which types of security issues (like phishing emails or malware infections) are serious enough to be considered 'actionable security incidents' – meaning they need to be dealt with. It also outlines what actions to take when these incidents happen.
- **How:** It provides the specific, step-by-step instructions on exactly how team members should carry out their assigned tasks.
- **When:** It details the exact circumstances or triggers that indicate it's time for team members to perform certain actions.

Incident response frameworks: Phases of incident response

Imagine you're dealing with a home emergency, like a burst pipe. You wouldn't just start randomly fixing things, right? You'd have a general idea of what needs to be done, step-by-step. In the world of computer security, 'incident response frameworks' are like pre-made emergency plans for dealing with cyber attacks.

Instead of figuring it all out from scratch when an attack happens, organizations can use these established frameworks. Think of them as guides that suggest a series of stages or 'phases' to follow. While different frameworks might have slightly different names or group things a little differently, they all cover similar core steps. These phases help ensure that responding to a security incident is a structured and organized process.

Here are the common phases of incident response:

1. **Preparation/Planning:** This is like getting your emergency kit ready before a storm. It involves building your incident response team (the people who will handle the problem), creating rules and guides ('policies' and 'processes'), and developing specific 'playbooks' (step-by-step instructions for common scenarios). You also set up the tools and services that will help your team during an incident.
2. **Detection/Identification:** This is like hearing the smoke alarm go off. It's about using IT monitoring systems (tools that watch over your computer systems) to spot, assess, confirm, and prioritize potential security incidents.
3. **Containment:** Once you know there's a fire, you need to stop it from spreading. In IT terms, this means taking action to prevent an incident from getting worse and to regain control over your computer systems.
4. **Eradication:** This is like putting out the fire. It involves removing the threat entirely, such as deleting malware (harmful software) or disabling malicious user accounts. You also figure out how the attackers got in ('vulnerabilities' – weaknesses in your systems) so you can fix them.
5. **Recovery:** After the fire is out, you need to rebuild and get back to normal. This means restoring normal operations and fixing any weaknesses that were exploited so the same problem can't happen again.
6. **Lessons Learned:** This is like reviewing what happened after the emergency. You look back to understand exactly what occurred, when, and how. You identify any security tools, rules, or procedures that didn't work as well as they should have and figure out how to improve them. This helps update your incident response plan so you're even better prepared next time.

Types of incident response teams

Just like a fire department has different specialized units (like ladder trucks or hazmat teams) for different kinds of emergencies, computer security also has different types of teams dedicated to responding to problems.

The raw content mentions three common types of teams that help computers when something goes wrong:

- **Computer incident response team (CIRT):** This is a team focused on dealing with unexpected computer problems.
- **Computer security incident response team (CSIRT):** This team is specifically designed to handle security-related problems on computers.
- **Computer emergency response team (CERT):** This is another type of team that springs into action when there's a computer emergency.

Why is an Incident Response Plan Important

Imagine your house has a fire alarm and a plan for what to do if it goes off – where to go, who to call, and how to get everyone out safely. An incident response plan is like that, but for computer security problems.

Having a good plan helps in a few key ways:

1. **Minimizing the Impact of Incidents:** When a security problem, or an **incident** (an unexpected problem in a computer system), happens, a plan helps everyone act fast and follow steps they already know. This is like having a clear evacuation route during a fire, which helps reduce the damage, save money, and protect the organization's reputation.
2. **Improving Incident Detection:** A good plan includes ways to constantly watch security systems and check them regularly. This is like having a security guard who is always looking for suspicious activity. It helps find potential problems, called **threats** (actions that could harm your computer systems), earlier so you can stop them before they cause a major issue, or **breach** (unauthorized access to data).
3. **Streamlining Communication:** Think of a busy emergency room. Everyone knows their role and how to talk to each other to help patients quickly. An incident response plan makes sure everyone involved – like the IT team, bosses, and other employees – knows what's happening and what to do. This stops confusion and helps fix the problem faster.
4. **Legal and Regulatory Compliance:** Some jobs and businesses have rules they must follow about keeping data safe and secure. Having a plan shows that the organization is serious about following these rules, which can help avoid fines or penalties.
5. **Building Resilience:** When everyone in an organization knows about possible security dangers and what to do if something goes wrong, the whole company gets stronger. It's like a community practicing for emergencies; they become better prepared to handle whatever comes their way.